

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

CampusCare – Campus Complaint Management System

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Project Type: Web-Based Campus Service and Complaint Management System

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1. Introduction

1.1 Purpose

CampusCare is a centralized web-based Campus Service and Complaint Management System designed to allow students to report campus-related problems and enable staff members and administrators to manage, assign, track, and resolve those complaints.

The system provides a structured platform for handling issues related to Information Technology (IT), maintenance, electrical services, laboratories, and other campus-related services.

The purpose of this SRS is to define the functional and non-functional requirements, system features, user roles, interfaces, architecture, data requirements, and constraints of the CampusCare system.

The document serves as a reference for the design, development, testing, and evaluation of the system.

1.2 Scope

CampusCare will provide a centralized platform for managing campus complaints from submission to resolution.

- Students can register/login, submit complaints, specify category, description and location, upload supporting attachments, receive a unique complaint ID, track complaint status, view complaint history, and provide feedback after resolution.
- Staff members can login, view complaints assigned to them, view complaint details, update complaint status, add resolution remarks, and mark complaints as resolved.
- Administrators can manage users, departments, complaint categories, view and assign complaints, monitor complaint progress, and generate/view reports.

The core workflow is: Authentication → Complaint Management → Assignment → Tracking → Resolution → Feedback → Administration/Reporting.

1.3 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
API	Application Programming Interface
Admin	Administrator
Complaint	A report submitted by a student regarding a campus-related problem
Complaint ID	Unique identifier assigned to each complaint
Staff	Campus employee responsible for handling assigned complaints
Department	Organizational unit responsible for a particular type of campus service
Status	Current state of a complaint
Resolution	Action taken to solve a complaint

Feedback	Response provided by a student after complaint resolution
Authentication	Process of verifying a user's identity
Authorization	Process of determining what a user is allowed to access

1.4 References

The primary reference is the CampusCare Initial Project Proposal & SRS Outline provided for this project.

2. Overall Description

2.1 Product Perspective

CampusCare is a web-based information system that acts as a centralized platform for campus complaint management. The system follows a modular three-tier architecture consisting of the Presentation Layer, Business Logic Layer, and Data Layer.

The Presentation Layer provides interfaces for students, staff members, and administrators. The Business Logic Layer contains application logic and REST APIs. The Data Layer provides persistent storage for users, complaints, departments, assignments, status history, and feedback.

2.2 Product Functions

- Authentication: login, logout, authentication, and role-based access control.
- Complaint Management: complaint submission, categorization, description, location, attachments, complaint ID generation, and viewing.
- Complaint Assignment: assignment of complaints to appropriate staff members.
- Complaint Tracking: viewing current complaint status and status history.
- Resolution: status updates, resolution remarks, and complaint closure.
- Feedback: student feedback after complaint resolution.
- Administration: management of users, departments, categories, complaints, and monitoring.
- Reporting: complaint-related reports and administrative monitoring.

2.3 User Classes

User	Responsibilities
Student/Faculty	Submit complaints, track complaints, view history, provide feedback.
Staff	View assigned complaints, update complaint status, add resolution remarks, resolve complaints.
Administrator	Manage users/departments/categories, assign complaints, monitor complaints, and manage reports.

2.4 Operating Environment

CampusCare is intended to operate as a web-based application accessible through commonly used modern web browsers.

- Google Chrome
- Microsoft Edge

- Mozilla Firefox
- Safari

The interface should support standard desktop and mobile screen sizes.

The server environment will include an application/server layer, REST API/business logic layer, database server, and storage for supporting attachments where required.

2.5 Constraints

1. Protected functionality requires authentication.
2. Users can only access functionality permitted by their assigned role.
3. Complaint assignment depends on available departments and staff members.
4. Complaint information must be stored consistently.
5. The system depends on a functioning database.
6. The system requires network access because it is web-based.
7. Supporting attachments may be subject to file type and size restrictions.
8. The system should support commonly used modern web browsers.

2.6 Assumptions and Dependencies

Assumptions

- Students, staff members, and administrators have valid accounts.
- Users provide accurate information while submitting complaints.
- Administrators maintain correct user, department, and category information.
- Staff members update complaint statuses accurately.
- The database remains available during normal system operation.
- Users have access to a compatible browser and network connection.

Dependencies

- Web browser
- Application/server environment
- Database system
- Network connectivity
- Authentication mechanism
- Storage mechanism for supporting attachments

3. System Features

3.1 Authentication

- Login and logout.
- User authentication.
- Role identification.
- Role-based access control.

3.2 Complaint Management

- Create complaints.

- Select complaint category.
- Enter description and location.
- Upload supporting files.
- Generate a unique complaint ID.
- View submitted complaints.

3.3 Complaint Assignment

- Administrator views complaints.
- Identify responsible department/staff.
- Assign complaints to staff.
- Track assignment information.

3.4 Complaint Tracking

- Students view current complaint status and history.
- Staff view assigned complaints.
- Administrators monitor complaint progress.

3.5 Resolution

- Staff review assigned complaints.
- Update complaint status.
- Enter resolution remarks.
- Mark complaints as resolved.

3.6 Feedback

- Students provide feedback after resolution.
- Feedback is associated with the corresponding complaint.

3.7 Administration

- Manage users.
- Manage departments.
- Manage complaint categories.
- View and monitor complaints.

3.8 Reporting

- View complaint-related reports.
- Monitor complaint activity.
- Provide administrative information for system monitoring.

4. Functional Requirements

ID	Requirement
FR-01	The system shall allow users to securely log in and log out using their registered credentials.
FR-02	The system shall provide role-based access to features based on whether the user is a student, staff member, or administrator.

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FR-03	If self-registration is enabled, the system shall allow new users to register using the required information.
FR-04	The system shall allow students to submit a complaint by providing its category, description, location, and supporting attachments.
FR-05	The system shall generate a unique complaint ID for every successfully submitted complaint.
FR-06	The system shall allow students to view their submitted complaints and track their current status.
FR-07	The system shall allow administrators to view complaints and assign them to appropriate staff members.
FR-08	The system shall allow assigned staff members to view complaints and update their status and resolution details.
FR-09	The system shall maintain the complete status history of each complaint from submission to closure.
FR-10	The system shall allow students to provide feedback for complaints that have been resolved.
FR-11	The system shall allow administrators to manage users, departments, complaint categories, and complaint reports.

5. Non-Functional Requirements

ID	Requirement
NFR-01 — Performance	The system shall respond to 95% of normal user requests within 2 seconds under the expected campus workload.
NFR-02 — Security	The system shall protect user credentials and restrict access to system functions according to the user's assigned role.
NFR-03 — Usability	The system shall provide a simple and intuitive interface that allows users to perform their required tasks with minimal training.
NFR-04 — Reliability	The system shall preserve complaint and user data accurately and prevent unintended loss or corruption of stored information.
NFR-05 — Availability	The system shall remain available to authorized users during the defined operating period, except during scheduled maintenance.
NFR-06 — Scalability	The system shall support the addition of new users, departments, complaint categories, and complaints without requiring major architectural changes.
NFR-07 — Maintainability	The system shall use modular components with clearly defined responsibilities so that individual components can be modified or maintained independently.

NFR-08 — Compatibility	The system shall function correctly on commonly used modern web browsers and standard desktop and mobile screen sizes.
NFR-09 — Data Integrity	The system shall ensure that each complaint, assignment, status update, and feedback record is stored consistently and associated with the correct user and complaint.
NFR-10 — Extensibility	The system shall be designed to allow future features, such as automated complaint categorization, notifications, analytics, or mobile application support, to be added with minimal changes to existing components.

6. External Interface Requirements

6.1 User Interface

Student/Faculty Interface

- Login
- Dashboard
- Submit Complaint
- My Complaints
- Complaint Details
- Complaint Status
- Complaint History
- Feedback

Staff Interface

- Login
- Staff Dashboard
- Assigned Complaints
- Complaint Details
- Status Update
- Resolution Remarks

Administrator Interface

- Login
- Admin Dashboard
- Users
- Departments
- Complaint Categories
- Complaints
- Assignment
- Monitoring
- Reports

6.2 API Interface

The Business Logic Layer shall provide REST APIs for communication between the user interface and backend services.

- Authentication API: /auth/login, /auth/logout, /auth/register
- Complaint API: /complaints, /complaints/{id}
- Assignment API: /assignments
- Tracking API: /complaints/{id}/history
- Feedback API: /complaints/{id}/feedback
- Administration API: /users, /departments, /categories, /reports

The exact API endpoints, HTTP methods, request formats, and response formats will be finalized during implementation.

6.3 Database Interface

The system shall use a persistent database for storing system information.

- Users
- Complaints
- Departments
- Assignments
- Complaint status history
- Feedback records

7. System Architecture

CampusCare follows a Modular Three-Tier Architecture.

- Presentation Layer
- Business Logic Layer / REST API
- Data Layer / Database

7.1 Presentation Layer

Provides the user interface for students, staff members, and administrators.

7.2 Business Logic Layer

Contains the main application logic and REST APIs. Modules include Authentication, Complaint Management, Assignment, Tracking, Feedback, and Administration.

7.3 Data Layer

Manages persistent data including users, complaints, departments, assignments, complaint status history, and feedback records.

8. Use Case Diagram

Primary actors:

- Student/Faculty (**Requester**)
- Staff
- Administrator

Student use cases: Login, Register (if enabled), Submit Complaint, Track Complaint, View History, Give Feedback.

Staff use cases: Login, View Assigned Complaints, Update Complaint, Resolve Complaints.

Administrator use cases: Login, View Complaints, Assign Complaints, Monitor Complaints, Manage Users, Manage Departments, Manage Complaint Categories, Generate/View Reports.

9. Detailed Use Cases

UC-01: Login

Actor: Student / Faculty / Staff / Administrator

Precondition: User has a registered account.

Main Flow:

9. User opens login page.
10. User enters credentials.
11. System validates credentials.
12. System identifies role.
13. System grants access to the appropriate dashboard.

Alternative Flow: Invalid credentials are rejected and an error message is displayed.

Postcondition: Authenticated user is redirected to the appropriate dashboard.

UC-02: Submit Complaint

Actor: Student/ Faculty

Precondition: Student/Faculty is authenticated.

Main Flow:

14. Student/Faculty opens Submit Complaint.
15. Student/Faculty selects category.
16. Student/Faculty enters description.
17. Student/Faculty enters location.
18. Student/Faculty attaches supporting files if required.
19. Student/Faculty submits complaint.
20. System validates information.
21. System creates complaint.
22. System generates unique complaint ID.
23. System displays confirmation.

Alternative Flow: Invalid or missing required information prevents successful submission.

Postcondition: Complaint is stored and available for further processing.

UC-03: Track Complaint

Actor: Student/Faculty

Precondition: Student has submitted a complaint.

Main Flow:

24. Student/Faculty opens My Complaints.
25. System displays submitted complaints.
26. Student selects a complaint.
27. System displays current status and available history.

Alternative Flow: If no complaints exist, the system displays an appropriate message.

Postcondition: Student/Faculty can determine the complaint's current state.

UC-04: View Assigned Complaints

Actor: Staff

Precondition: Staff member is authenticated.

Main Flow:

28. Staff opens assigned complaints.
29. System retrieves complaints assigned to the staff member.
30. System displays complaint information.
31. Staff selects a complaint to view details.

Alternative Flow: Unassigned complaints are not shown in the staff member's assigned list.

Postcondition: Staff can access assigned complaint information.

UC-05: Update Complaint

Actor: Staff

Precondition: Complaint is assigned to the staff member.

Main Flow:

32. Staff opens assigned complaint.
33. Staff reviews complaint information.
34. Staff updates status.
35. Staff enters remarks.
36. System stores update.
37. System updates status history.

Alternative Flow: Unauthorized staff cannot update a complaint outside their permitted scope.

Postcondition: Updated complaint information is stored.

UC-06: Resolve Complaint

Actor: Staff

Precondition: Complaint has been assigned to staff.

Main Flow:

38. Staff reviews complaint.
39. Staff performs required resolution.
40. Staff enters resolution details.
41. Staff updates status.
42. System records resolution.
43. System updates history.

Alternative Flow: Resolution cannot be recorded without the required resolution information.

Postcondition: Complaint is marked as resolved according to the workflow.

UC-07: Assign Complaint

Actor: Administrator

Precondition: Complaint has been submitted.

Main Flow:

44. Administrator opens complaint list.
45. Administrator selects complaint.
46. Administrator identifies department/staff.
47. Administrator assigns complaint.
48. System stores assignment.
49. Assigned staff can view complaint.

Alternative Flow: Invalid or unavailable assignment targets cannot be selected.

Postcondition: Complaint is associated with selected staff member.

UC-08: Give Feedback

Actor: Student/Faculty

Precondition: Complaint has been resolved.

Main Flow:

50. Student/Faculty opens resolved complaint.
51. Student/Faculty selects feedback option.
52. Student/Faculty provides feedback.
53. Student/Faculty submits feedback.
54. System stores feedback.

55. Feedback is associated with complaint.

Alternative Flow: Feedback is only available when the complaint is eligible for feedback.

Postcondition: Feedback is stored successfully.

UC-09: Manage Users

Actor: Administrator

Precondition: Administrator is authenticated.

Main Flow:

56. Administrator opens user management.
57. System displays users.
58. Administrator selects a user.
59. Administrator performs an allowed operation.
60. System stores updated information.

Alternative Flow: Unauthorized users cannot access user management.

Postcondition: User information is updated.

UC-10: Monitor Complaints

Actor: Administrator

Precondition: Administrator is authenticated.

Main Flow:

61. Administrator opens complaint monitoring.
62. System retrieves complaint information.
63. Administrator views statuses.
64. Administrator monitors progress.
65. Administrator identifies pending or unresolved complaints.

Alternative Flow: Unavailable data is handled with an appropriate message.

Postcondition: Administrator has an overview of complaint activity.

10. Data Requirements

10.1 User

- User ID
- Name
- Email
- Password/Credential Information
- Role
- Department

- Account Status

10.2 Complaint

- Complaint ID
- Student/User ID
- Category ID
- Description
- Location
- Attachment
- Submission Date
- Current Status

10.3 Department

- Department ID
- Department Name
- Description
- Status

10.4 Assignment

- Assignment ID
- Complaint ID
- Staff ID
- Department ID
- Assigned Date
- Assignment Status

10.5 Complaint Status History

- History ID
- Complaint ID
- Previous Status
- New Status
- Updated By
- Updated Date
- Remarks

10.6 Feedback

- Feedback ID
- Complaint ID
- Student ID
- Feedback
- Submitted Date

10.7 Conceptual Relationships

User submits Complaint. Complaint belongs to a Category, is associated with an Assignment, has Status History, and may have Feedback. Assignment connects a Complaint with a Staff member and Department. Users may belong to a Department.

11. System Constraints

66. Only authenticated users can access protected functionality.
67. Access must be controlled according to user role.
68. Students should only be able to manage/view their own complaints.
69. Staff members should only be able to access complaints assigned to them, according to the defined authorization rules.
70. Administrators have broader management and monitoring privileges.
71. Complaint records must remain associated with the correct user.
72. Assignment records must remain associated with the correct complaint and staff member.
73. Status history must remain associated with the corresponding complaint.
74. Feedback must remain associated with the corresponding complaint and student.
75. The system requires persistent data storage.
76. The system requires network connectivity for normal web access.
77. The system should support commonly used modern web browsers.

12. Acceptance Criteria

Authentication

- Registered users can log in using valid credentials.
- Invalid credentials are rejected.
- Users receive access according to their role.
- Users can log out successfully.

Complaint Submission

- Student/Faculty can select category.
- Student/Faculty can enter description.
- Student/Faculty can provide location.
- Student/Faculty can upload supporting attachments where applicable.
- Required information is validated.
- Successful complaints receive a unique complaint ID.

Complaint Tracking

- Student/Faculty can view submitted complaints.
- Student/Faculty can view current status.
- Student/Faculty can view available complaint history.
- Student/Faculty cannot access another student's complaints.

Complaint Assignment

- Administrator can view submitted complaints.
- Administrator can assign a complaint to an appropriate staff member.
- Assignment information is stored with the complaint.

Complaint Update

- Assigned staff can view assigned complaints.
- Staff can update complaint status.

- Staff can enter resolution remarks.
- Status changes are recorded in history.

Complaint Resolution

- Staff can mark a complaint as resolved according to the workflow.
- Resolution information is stored.
- Updated status is visible to the student.

Feedback

- Student/Faculty can provide feedback after resolution.
- Feedback is associated with the correct complaint.
- Feedback is stored successfully.

Administration

- Administrator can manage users.
- Administrator can manage departments.
- Administrator can manage complaint categories.
- Administrator can view/monitor complaints.
- Administrator can access complaint reports.

13. Future Enhancements

13.1 Automated Complaint Categorization

Machine-learning-based classification could automatically suggest an appropriate complaint category based on the complaint description.

13.2 Notifications

The system could provide email notifications, in-app notifications, status-change notifications, and assignment notifications.

13.3 Analytics Dashboard

The system could provide graphical analytics such as complaint trends, department-wise complaints, resolution time, category-wise distribution, and pending complaint analysis.

13.4 Mobile Application

A dedicated mobile application could be developed for students, staff, and administrators.

13.5 Advanced Reporting

Additional reporting could include monthly reports, department performance reports, complaint resolution statistics, and historical trend analysis.

14. Summary of System Workflow

Student submits complaint → System generates complaint ID → Administrator reviews and assigns complaint → Staff views assigned complaint → Staff updates complaint → Staff resolves complaint → System updates status/history → Student views resolution → Student provides feedback.

15. Requirement Traceability Summary

Module	Main Requirements
Authentication	FR-01, FR-02
Registration	FR-03, if enabled
Complaint Management	FR-04, FR-05, FR-06
Assignment	FR-07
Staff Management	FR-08
Tracking	FR-09
Feedback	FR-10
Administration	FR-11
Performance	NFR-01
Security	NFR-02
Usability	NFR-03
Reliability	NFR-04
Availability	NFR-05
Scalability	NFR-06
Maintainability	NFR-07
Compatibility	NFR-08
Data Integrity	NFR-09
Extensibility	NFR-10